An Extensible Personal Finance Management Framework for Expense Tracking, Visual Analytics, and Intelligent Decision Support

COHORT

Networks, Security, Forensic and Blockchain

INNOVATION

Automated Personal Finance Management

RESEARCH

Expense Visualization and Spending Pattern Analysis

IMPACT

Data-Driven Financial Planning and Decision Making

Abstract

Managing personal finances effectively has become increasingly challenging in our day-to-day lives, as we make frequent payments through UPI, cards, cash, online shopping, subscriptions, and other digital platforms. Although existing finance applications help users record these transactions, they often provide limited support for understanding spending habits, identifying overspending, and making better financial decisions.To address this problem, our project, “An Extensible Personal Finance Management Framework for Expense Tracking, Visual Analytics, and Intelligent Decision Support,” proposes a more integrated approach to personal finance management. The system allows users to record and organize their income and expenses while using visual dashboards to make spending patterns easier to understand. It also focuses on identifying spending trends, monitoring budgets, and providing personalized insights and alerts based on user behavior.The framework is designed to be simple, flexible, and extensible, allowing additional features such as spending prediction, anomaly detection, and advanced recommendations to be incorporated in the future. Overall, the project aims to move beyond basic expense recording and help users understand their financial behavior, recognize areas of overspending, and make more informed financial decisions.

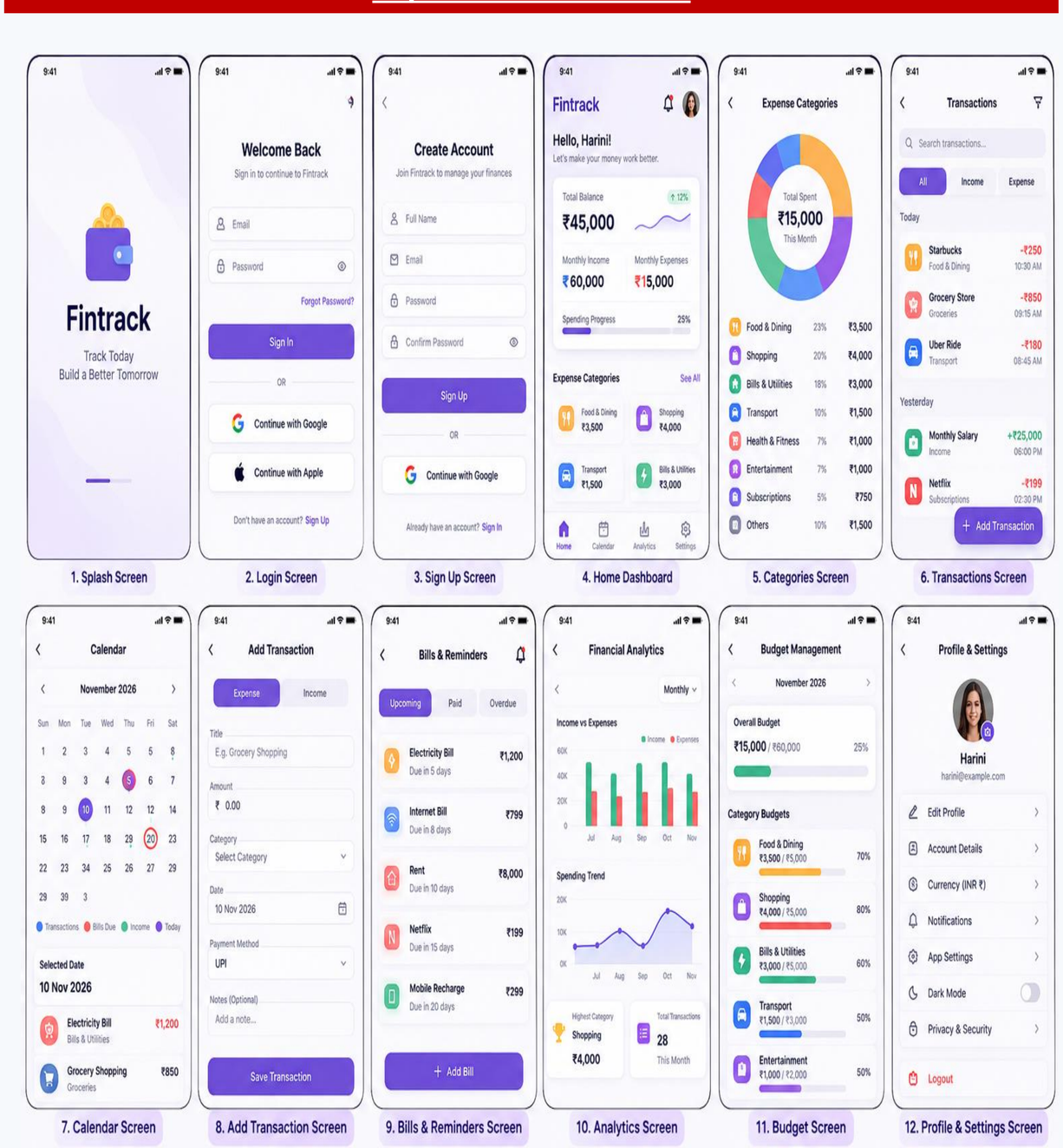
Proposed Methodology



Objectives

- To develop a user-friendly framework for systematic income and expense tracking.
- To integrate visual analytics and spending-pattern analysis for meaningful financial insights.
- To enhance traditional expense tracking with personalized budgeting, alerts, and intelligent decision support.
- To develop an extensible modular architecture for incorporating future financial analytics and recommendation features.
- To evaluate the proposed framework based on usability, analytical accuracy, visualization effectiveness, and usefulness of financial insights.

Experimental Results



References

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